

**Nirma University**  
Institute of Technology  
Semester End Examination (IR), January-2025  
M.C.A, Semester- I  
6CS101CC22 Object Oriented Programming

Roll/  
Exam No.

Supervisor's initial  
with date

**Time: 3 Hours**

**Max. Marks: 100**

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

**SECTION-I**

- Q-1** **Answers the following:** [18]  
(A) Explain the different features of the object-oriented programming. (6)  
CO1,BL2  
(B) Differentiate between the “while” and “do while” loop by giving (6)  
CO1,BL4 suitable example.  
(C) Briefly describe JDBC (Java Database Connectivity) and JDBC (6)  
CO3,BL2 Drivers. Explain the use of DriverManager class with an example.
- Q-2** **Answers the following:** [16]  
(A) What is the function of the Java Development Kit (JDK) in Java (8)  
CO2,BL4 programming? Discuss its components in detail.  
(B) Explain the role of the static and non-static variables by giving (8)  
CO2,BL5 a suitable example.
- Q-3** **Answers the following:** [16]  
(A) Discuss the “String” class. Explain the compareTo(), contains() (8)  
CO2,BL3 and substring() methods of String class with a suitable example.  
(B) How Java constructor differs from Java method? Explain by (8)  
CO2,BL4 giving a suitable example.

**SECTION-II**

- Q-4** **Answers the following:** [18]  
(A) What is the use of Thread class in Java programming? Develop (6)  
CO3,BL6 a java program which illustrates the usage of the Thread class.  
(B) What are the advantages of creating package in java (6)  
CO3,BL3 programming? How do you create package and import it into Java program?  
(C) Discuss the function of an “interface” in java programming. (6)  
CO2,BL6 Support your answer by giving a suitable example.

- Q-5** **Answers the following:** [16]
- (A) How run-time polymorphism differs from compile-time polymorphism? Write a java program to demonstrate the compile-time polymorphism. (8)  
CO2,BL4
- (B) Discuss the “private” and “public” access modifiers by giving a suitable example of java program. (8)  
CO2,BL6
- Q-6** **Answers the following:** [16]
- (A) What is the role of the file management in developing java program? Write a java program which creates a file and stores the data in the file entered by the user through command prompt. (8)  
CO3,BL5
- (B) Identify the use of exception handling in Java programming. (8)  
CO3,BL3
- Explain the try and catch block with a suitable example which uses an exception `ArrayIndexOutOfBoundsException` in java program.